Introduction to Probability and Statistics

(EE 354 / CE 361)

PMFs, Expected Value, Variance, Standard Deviation

Today's Agenda

Mean & PMF of :

- Binomial Distribution
- Bernoulli Distribution
- Discrete Uniform Distribution
- Geometric Distribution

Practice Problems

1. $\mu_X = E[X] = \sum_x x \cdot p_X(x)$ *

2. Expected Value Rule

$Y = g(X)$

$$\mu_Y = E[Y] = E[g(X)] = \sum_x g(x) p_X(x)$$

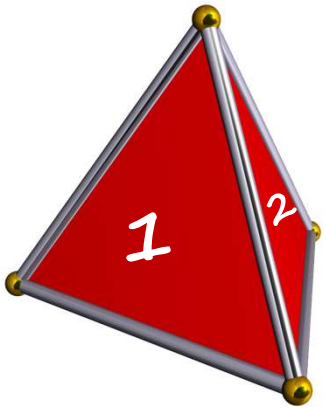
3. $\text{var}(X) = E[(X - E[X])^2]$

$f(x) = (X - E[X])^2$

$$\text{var}(X) = \sum_x (x - E[X]^*)^2 p_X(x)$$

↳ Constant $= \mu_X$

Exercise # 1

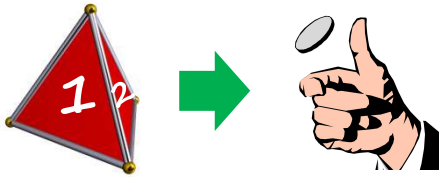


Equally likely

- You throw a tetrahedral dice 12 times.

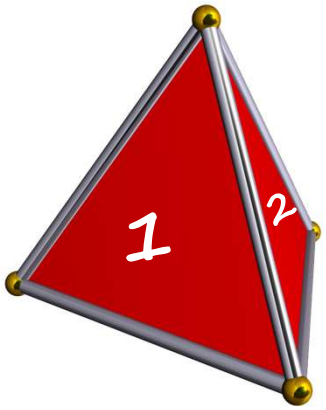
R.V X is the number of times you get 3 in the 12 throws & $P_X(k)$ is the Probability of getting No 3, k-times in 12 throws

What is $P_X(5)$ =



parameters $p = 1/4$

Exercise # 1

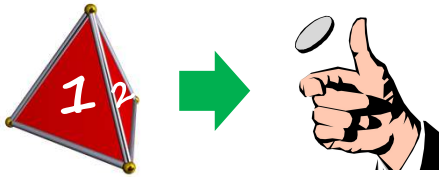


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parameters p = 1/4

N = 12, p = ¼ , 1-p = ¾ , k=5

$$p_X(5) = \binom{12}{5} \cdot \left(\frac{1}{4}\right)^5 \cdot \left(\frac{3}{4}\right)^7$$

Exercise No 02!

A graduating CS student of HU goes to career fair booths in technology sector - & his/her likelihood of receiving an off-campus interview invitation after a career fair booth visit depends on how well he/she did in EE 354 course. Specifically, an A in the course results in a probability $p_1 = 0.95$ of obtaining an invitation, whereas a C in EE 354 course results in a probability of $p_2 = 0.15$ of an invitation. Assuming that each student visits 5 booths during a typical career fair, find the probability that an A grade student in EE 354 course will not get an off-campus interview invitation. Similarly, find the probability that a C grade student in EE 354 course will get an invitation during a typical career fair

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P(an A grade student does not get an invitation in 5 trials)= $\sum_{k=6}^{\infty}(1-p_1)^{k-1.}p_1 = (1-p_1)^5$

P(C grade student gets an invitation in 5 trials)= $1-(1-p_2)^5$

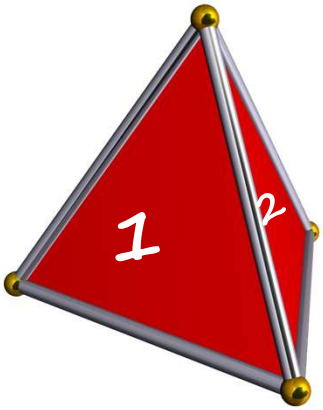
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P(an A grade student does not get an invitation in 5 trials)= $\sum_{k=6}^{\infty}(1-p_1)^{k-1.}\,p_1 = (1-p_1)^5$
$= (1-0.95)^5 = (0.05)^5$

P(C grade student gets an invitation in 5 trials)= $1-(1-p_2)^5 = (1-0.15)^5 = (0.85)^5$

Exercise # 3



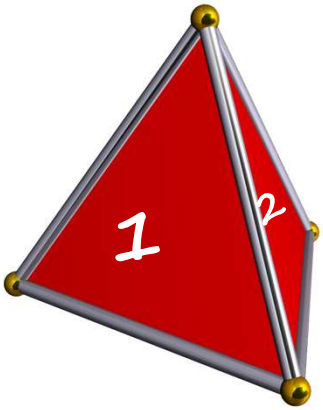
Equally likely

A Random Variable Z_n is defined as $Z_n = \min(X_1; X_2; \ldots; X_n)$, where $X_{i's}$ are outcomes of identical and independently distributed tetrahedral die (4-sided). As an example $Z_2 = \min(X_1;X_2)$ - meaning Z_2 is the minimum of the outcome of two independent throws of tetrahedral die X_1 and X_2.

1. What is the $P(Z_{10} > 1)$?

2. What is the $P(Z_{10} = 3)$?

Exercise # 3



Equally likely

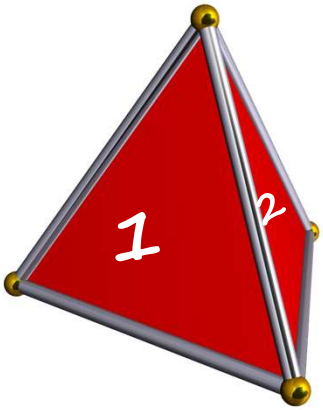
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1. What is the $P(Z_{10} > 1)$?

$$P(Z_{10} > 1) = \left(\frac{3}{4}\right)^{10}$$

2. What is the $P(Z_{10} = 3)$?

Exercise # 3



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1. What is the $P(Z_{10} > 1)$?

$$P(Z_{10} > 1) = \left(\frac{3}{4}\right)^{10}$$

We define two events
A - Either 3 or 4 appears in ALL ten throws,
B - 4 appears in ALL ten throws

2. What is the $P(Z_{10} = 3)$?

$$P(A) = \left(\frac{2}{4}\right)^{10} \qquad P(B) = \left(\frac{1}{4}\right)^{10}$$

$$P(Z_{10} = 3) = \left(\frac{2}{4}\right)^{10} - \left(\frac{1}{4}\right)^{10}$$

Exercise # 4

A university IT helpdesk receives support requests each hour. The random variable X (can take on 5 values that represents the number of requests received in an hour, with the following probabilities:

P(X=o) = 2p, P(X=1) = 3P, P(X=2) = 2.5P, P(X=3) = 1.5p, P(X=4) = p

(a) Calculate p and plot the probability mass function for the random variable X.

The helpdesk has 2 staff members, and each can handle at most 1 request per hour. If more than 2 requests arrive, the extra requests are left unresolved. Define a new random variable Y = number of unresolved requests in an hour.

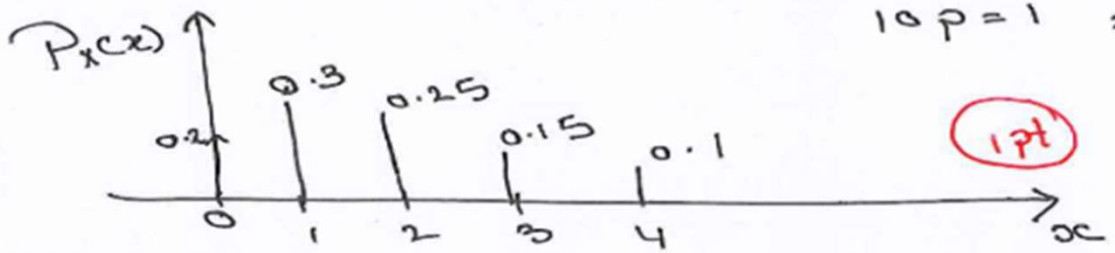
(b) Write down and plot the probability mass function for random variable Y.

(c) What is the probability that all requests in an hour cannot be resolved?

(d) Compute E[Y], the expected number of unresolved requests per hour.

(a) for a valid distribution, $2p + 3p + 2.5p + 1.5p + p = 1$

$10p = 1 \Rightarrow p = 0.1$ (2pts)



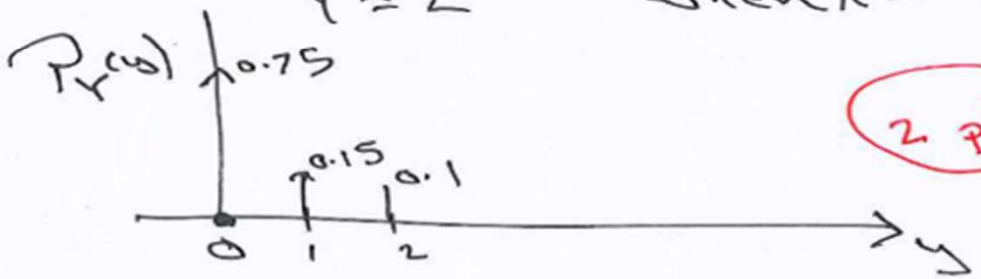
(1pt)

(b) Possible values for RV $Y = \{0, 1, 2\}$ (1pts)

$Y = 0$ when $X = 0, 1, 2 \Rightarrow P(Y=0) = 0.75$

$Y = 1$ when $X = 3 \Rightarrow P(Y=1) = 0.15$

$Y = 2$ when $X = 4 \Rightarrow P(Y=2) = 0.1$ (1pt)



(2 pt)

(c) Probability that all requests cannot be resolved (2pts)

$\Rightarrow Y \geq 1 \Rightarrow P(Y=1 \text{ or } Y=2)$

$= P(Y=1) + P(Y=2) = 0.15 + 0.1$

$= 0.25$ (2 pts)

(d) $E[Y] = \sum_y y P_Y(y)$ (1pt)

$= 0 \times 0.75 + 1 \times 0.15 + 2 \times 0.1$ (1pt) (1pt)

$= 0.15 + 0.2 = 0.35$

Exercise # 5

Let S be a set of integers and $g(x)$ defined as :

$$g(x) = \begin{cases} k(7x+3) & x = 1, 2, 3...7 \\ 0 & otherwise \end{cases}$$

For what value of k is $g(x)$ a valid PMF?

Exercise # 5

Let S be a set of integers and $g(x)$ defined as :

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For what value of k is $g(x)$ a valid PMF?

Solution 1

For a valid PMF first we know that $f(x) \geq 0$, so k is non-negative. Secondly, we know that $\sum_{x \in S} f(x) = 1$. Therefore:

$$\sum_{x=1}^{7} f(x) = 1$$

$$f(1) + f(2) + f(3) + f(4) + f(5) + f(6) + f(7) = 1$$

$$k(7+3) + k(14+3) + k(21+3) + k(28+3) + k(35+3) + k(42+3) + k(49+3) = 1$$

$$10k + 17k + 24k + 31k + 38k + 45k + 52k = 1$$

$$217k = 1$$

$$k = \frac{1}{217}$$

Exercise # 6

Determine the constant c for which the random variable Y is a valid PMF.

$$f(y) = \begin{cases} c(\frac{1}{4})^y & y = 1, 2, 3, \ldots \\ 0 & otherwise \end{cases}$$

Solution 2

For a valid PMF first we know that $f(y) \geq 0$, so c is non-negative. Secondly, we know that $\sum_{y=1}^{\infty} f(y) = 1$. Therefore:

$$\sum_{y=1}^{\infty} f(y) = 1$$

$$\sum_{y=1}^{\infty} c(\frac{1}{4})^y = 1$$

$$c \times \sum_{y=1}^{\infty} (\frac{1}{4})^y = 1$$

$$c \times \left(\frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \frac{1}{4^4} + ... \right)$$

It can be seen that the integers follow a geometric series so we can use the formula for sum to infinity :

$$S = \frac{a}{1-r} \text{ where } a = \frac{1}{4}, r = \frac{1}{4}$$

$$c \times \left(\frac{a}{1-r} \right) = 1$$

$$c \times \left(\frac{\frac{1}{4}}{1-\frac{1}{4}} \right) = 1$$

$$c \times \left(\frac{\frac{1}{4}}{\frac{3}{4}} \right) = 1$$

$$c \times \frac{1}{3} = 1$$

$$c = 3$$

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